

- 5 The causative organism of measles is *Morbillivirus*. Young children who have not been vaccinated for measles are at highest risk of becoming ill and of developing complications associated with the disease.

(a) The genetic material of *Morbillivirus* is a strand of RNA.

Statements **A**, **B** and **C** relate to the structure of RNA.

State the correct term or terms to match each of statements **A** to **C**.

A The names of the two purines and two pyrimidines in RNA.

purines

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pyrimidines

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B The type of covalent bond between RNA nucleotides.

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C The pentose sugar of the RNA nucleotide.

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[4]

(b) Describe how measles is transmitted.

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..... [3]

- (c) Young children with measles may develop difficulties with breathing. This is made worse if the child is continually exposed to tobacco smoke. Tobacco smoke contains harmful compounds, such as carbon monoxide.
- (i) Inhaled tobacco smoke passes through the larynx (voice box) and other structures in the gas exchange system to reach the alveoli.

Complete Fig. 5.1 to list, in the correct sequence, the main structures of the gas exchange system through which the tobacco smoke passes to reach the alveoli.



Fig. 5.1

[1]

- (ii) State the short-term effects of carbon monoxide on the cardiovascular system.

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..... [2]

- (d) Smallpox is a disease that has been eradicated with the help of a global vaccination programme. Measles has not been eradicated, even though there is a global vaccination programme.

Fig. 5.2 is a graph showing the estimates of number of deaths from measles globally, between 2000 and 2017, for people who have **not** been vaccinated.

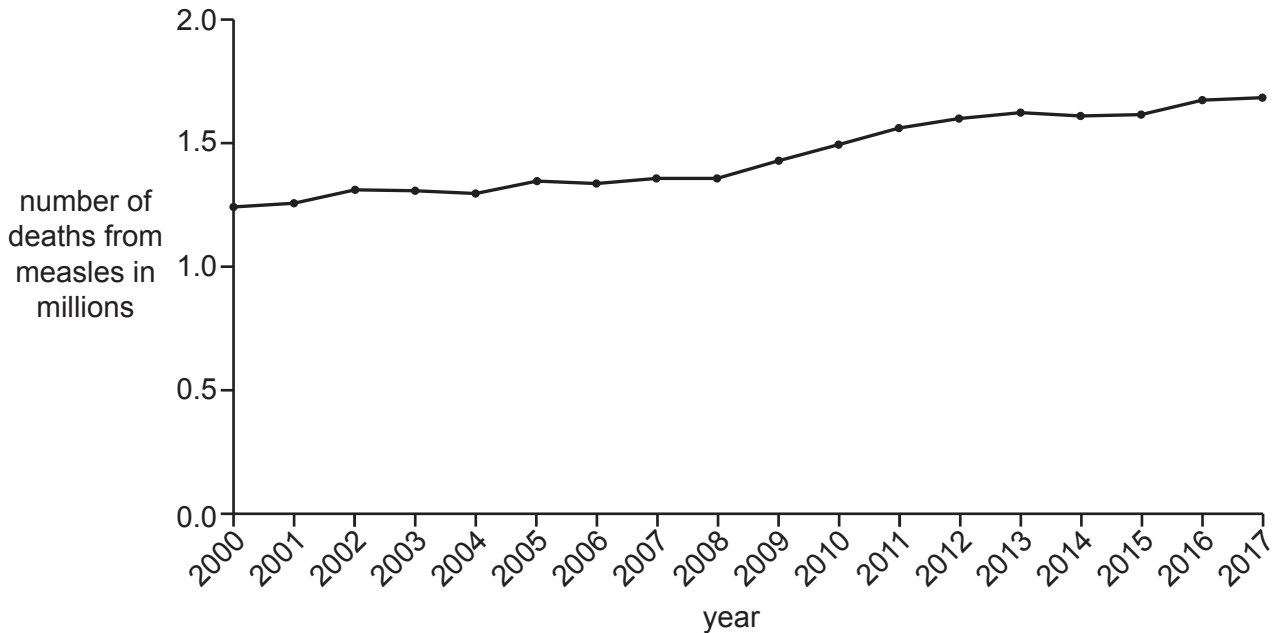


Fig. 5.2

- (i) State the trend shown in Fig. 5.2 between 2000 and 2017.

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..... [1]

(ii) Suggest reasons for the trend shown in Fig. 5.2.

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..... [3]

(e) A child with leukaemia is at high risk of developing measles even though the child has a high white blood cell count **and** has been vaccinated against the disease.

Explain why this child is at high risk of developing measles.

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..... [2]

[Total: 16]